

QATIPv3 Azure Lab3

VNet deployment using Terraform

Contents

Lab Objectives.....	1
Teaching Points	2
Before you begin.....	2
Solution.....	2
Task 1. Create the Resource group, VNet and Subnet	2
Task 2. Create Routing Table and Routing Table Association.....	6
Task 3. Create Network Security Group.....	8
Task 4. Create Virtual Machine with Virtual Nic and Public IP	10
Task 5. Testing	10
Task 6. Lab Clean-up	11

Lab Objectives

1. In this lab, you will create the following resources...

Resource Group - “**RG3**” in West Europe

VNet - “**Lab3-Vnet**” using 10.1.0.0/16

Subnet - “**Lab3-Subnet**” using 10.1.1.0/24

Routing Table - “**Lab3-Route-Rable**” routing 0.0.0.0 traffic to the internet

Routing Table Association – between “**Lab3-Subnet**” and “**Lab3-Route-Table**”

NSG - “**WebAdminNSG**” allowing inbound RDP 3389 traffic

Security group association – between “**Lab3-Subnet**” and “**WebAdminNSG**”

Virtual machine – “**VM3**”

Virtual Nic - “**Lab3-Nic**”

Public IP - “**Lab3-Public-IP**”

This lab breaks a deployment into phases, each a separate task, to demonstrate component interdependencies:

Task 1: Resource Group, VNet and subnet

Task 2: Routing Table and Routing Table Association

Task 3: Network Security Group and Network Security Group association

Task 4: Virtual Machine with Virtual Nic and Public IP

Task 5: Testing

Teaching Points

This lab will take you through coding a multi-component cloud deployment into Azure using Terraform. You will deploy a virtual machine into a relatively simple network but even this has several components that need either to exist already or be created as part of the deployment. You have the option of attempting the tasks yourself with minimal guidance or following step-by-step instructions.

Before you begin

1. Ensure you have completed Lab0 before attempting this lab
2. In the IDE terminal pane, enter the following command... `cd c:\azurelabs\lab3`
3. This shifts your current working directory to lab3. Ensure all commands are executed in this directory
4. Close any open files and use the Explorer pane to navigate to and open the lab3 folder

Solution

The solution to this lab is in folder C:\azurelab\solutions\lab3. Try to use this only as a last resort if you are struggling to complete the step-by-step processes.

Task 1. Create the Resource group, VNet and Subnet

Try it yourself

Searching the Registry for guidance....

1. Add and configure an `azurerm_provider` block into `main.tf` using version **4.16.0**
2. Create an `azurerm_resource_group` resource with identifier **RG_3**. Name the resource group **RG3**. Specify **West Europe** as the location.
3. Create an `azurerm_virtual_network` resource with identifier **Lab3_Vnet**. Name the network **Lab3-Vnet**. Do not provision inline subnets or DNS Servers. Use a **10.1.0.0/16** address space and reference the name and location of the `azurerm_resource_group` you created in step 2.
4. Create an `azurerm_subnet` resource with identifier **Lab3_Subnet**. Name the subnet **Lab3-Subnet**. Delegation is not required. Use a **10.1.1.0/24** address space and reference the name and location of the `azurerm_resource_group` you created in step 2.
5. Deploy these resources and validate their creation through the Azure Portal

Step-by-Step

1. Open Terraform Registry Resources link [here](#)
2. Scroll down to the 'Example Usage' section and copy the sample code into the Task1 section of `main.tf`
3. Change the version value to **4.16.0**
4. Remove the `resource_provider_registrations` line
5. Add the following line under 'features {}', replacing <your subscription id here> with your Azure lab subscription id..

`subscription_id = "<your subscription id here>"`

6. Change the identifier of the `azurerm_resource_group` resource block from 'example' to **RG_3**
7. Change the name of the resource group to **RG3**
8. Change the `azurerm_virtual_network` resource block identifier from 'example' to **Lab3_Vnet**

9. Change the `azurerm_virtual_network` resource name from 'example-network' to **Lab3-Vnet**

10. Change the location value to **`azurerm_resource_group.RG_3.location`**

11. Change the `resource_group_name` to **`azurerm_resource_group.RG_3.name`**

These 2 settings link the creation of this network to the resource group.

Terraform will therefore create the resource group before creating the network

12. Change the virtual network address space to **`10.1.0.0/16`**

13. We will now add a subnet to the network. Open the Terraform Registry Resources link [here](#)

14. Scroll down to the 'Example Usage' section and copy just the 'azurerm_subnet' example block into `main.tf`, below your existing Task1 code.

15. Change the `azurerm_subnet` resource block identifier to "**Lab3_Subnet**"

16. Change the `azurerm_subnet` resource name to "**Lab3-Subnet**"

17. Change the `resource_group_name` to **`azurerm_resource_group.RG_3.name`**

18. Change the `virtual_network_name` value to **`azurerm_virtual_network.Lab3_Vnet.name`**

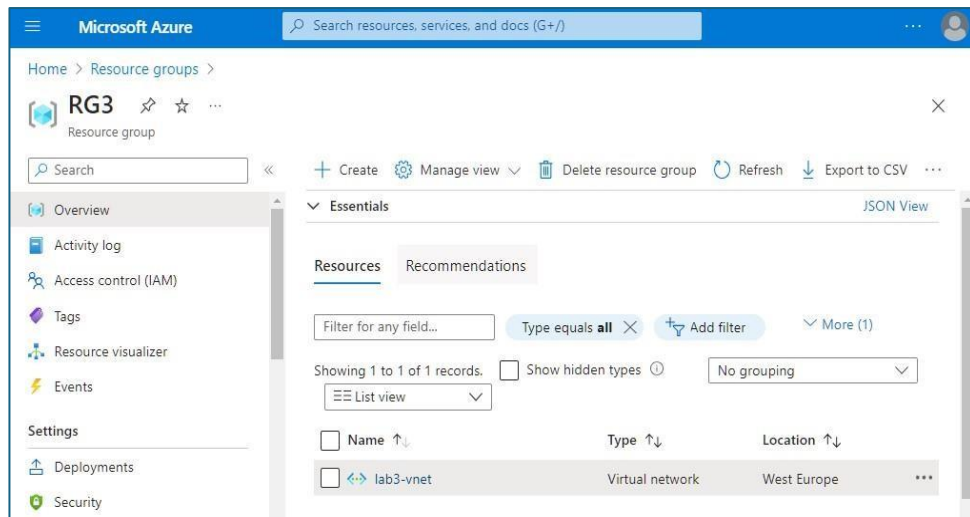
This links the subnet Lab3-Subnet to the network Lab3-Vnet which is part of resource group RG3. The subnet block references the network block, which references the resource group block. These references indicate the order that terraform should create the resources, resource group first, then network and finally the subnet. This is an example of implicit dependency.

19. Change the subnet address space to **`10.1.1.0/24`**

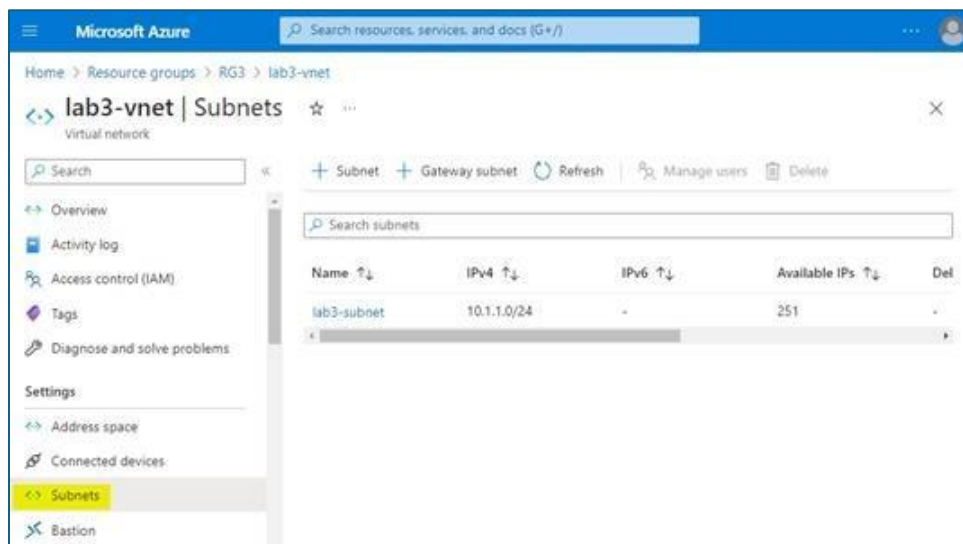
20. Delete the entire **delegation** sub-block

21. Initialize terraform with **`terraform init`**

22. Deploy these resources using **terraform plan** followed by **terraform apply**. Enter **yes** when prompted.
23. Switch to the Azure Portal. Verify the creation of resource group **RG3** and 2 associated resources, the vnet with a single subnet...



24. Click into **Lab3-Vnet** to verify the creation of the subnet...



Task 2. Create Routing Table and Routing Table Association

Try it yourself

1. Create an **azurerm_route_table** resource with identifier **Lab3_Route_Table**. Name the route table **Lab3-Route-Table**. Reference your **azurerm_resource_group** name and location. Configure local routing for all 10.1.0.0/16 destinations and the Internet for all others.
2. Create an **azurerm_subnet_route_table_association** resource with identifier **Lab3_Subnet_RTA**. Associate the **azurerm_subnet** and **azurerm_route_table** resources you created in Task 1.
3. Deploy these resources and validate their creation through the Azure Portal

Step-by-Step

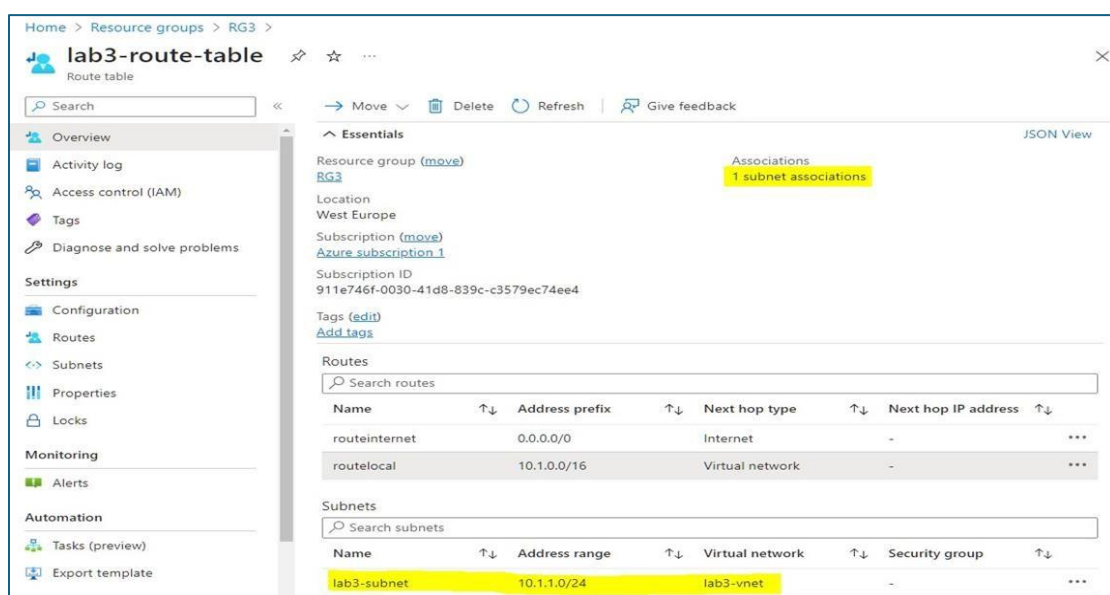
1. Open Terraform Registry Resources link [here](#)
2. Familiarize yourself and then copy the **azurerm_route_table** example block into the Tsk2 section of main.tf excluding the resource group resource block.
3. Change the resource block identifier to "**Lab3_Route_Table**"
4. Change the name to "**Lab3-Route-Table**"
5. Change the location to **azurerm_resource_group.RG_3.location**, thus referencing the location attribute of our resource group
6. Change the resource_group_name to **azurerm_resource_group.RG_3.name**, thus referencing the name attribute of our resource group
7. Rename the route to "**Route-Local**" Note that the address_prefix is correct and aligns to the vnet address space defines in Task 1.
8. Remove the **tags** sub-block
9. Add a second route block to add a default route to the internet...

```

route {
  name = "Route-Internet"
  address_prefix = "0.0.0.0/0"
  next_hop_type = "Internet"
}

```

10. We now need to associate our subnet with the route table. Open Terraform Registry Resources link [here](#)
11. Copy the last resource block example, 'azurerm_subnet_route_table_association', into the bottom of the Task 2 block of main.tf
12. Change the resource block identifier from 'example' to "Lab3_Subnet_RTA"
13. Change the subnet_id value to **azurerm_subnet.Lab3_Subnet.id**
14. Change the route_table_id to **azurerm_route_table.Lab3_Route_Table.id**
15. Deploy these resources using **terraform plan** followed by **terraform apply**
16. Switch to the Azure Portal. Verify the creation of 2 additional resource in the RG3 resource group...
17. Click into **lab3-route-table** to verify the creation of the subnet association...



Task 3. Create Network Security Group

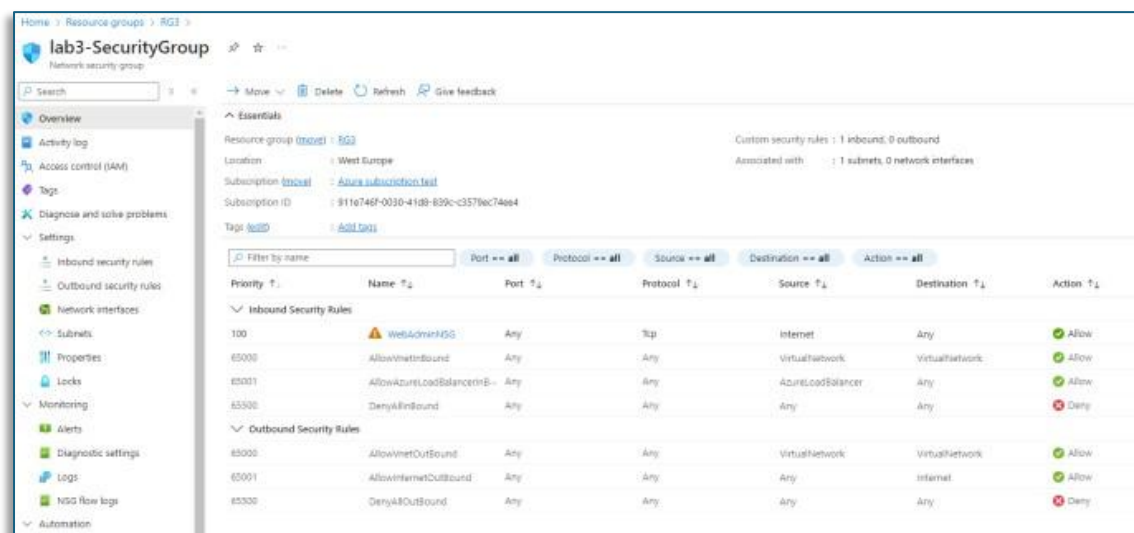
Try it yourself

1. Create an **azurerm_network_security_group** resource with identifier **Lab3_Security_Group**. Name the NSG **Lab3-Security-Group**. Reference it to your resource group name and location. Configure the security rule to permit inbound traffic from the internet trying to reach any destination on TCP port 3389.
2. Associate this NSG with your Vnet subnet using an **azurerm_subnet_network_security_group_association** resource with identifier **Lab3_NSG_Assoc**.
3. Deploy these resources and validate their creation through the Azure Portal.

Step-by-Step

1. Open Terraform Registry Resources link [here](#)
2. Familiarize yourself and then copy the **azurerm_network_security_group** example block into Task3 section of main.tf
3. Change the resource block identifier to **"Lab3_Security_Group"**
4. Change the resource name to **"Lab3-Security-Group"**
5. Change the location value to **azurerm_resource_group.RG_3.location**
6. Change the resource_group_name to **azurerm_resource_group.RG_3.name**
7. Change the security rule name to **"WebAdminNSG"**
8. Change the destination_port_range to **"3389"**
9. Change the source_address_prefix to **"Internet"**
10. Remove the **tags** sub-block

11. We must now associate this security group with our subnet. Open Terraform Registry Resources link [here](#)
12. Familiarize yourself and then copy the 'azurerm_subnet_network_security_group_association' sample block into the bottom of the Task 3 block in main.tf
13. Change the resource block identifier from 'example' to 'Lab3_NSG_Assoc'
14. Change the subnet_id to **azurerm_subnet.Lab3_Subnet.id**
15. Change the network_security_group_id to **azurerm_network_security_group.Lab3_Security_Group.id**
16. Deploy these resources using **terraform apply** followed by **yes**
17. Switch to the Azure Portal. Verify the creation of 2 additional resource in the RG3 resource group...
18. Click into the **lab3-SecurityGroup** network security group to verify the security rule and subnet association are created... The warning icon against the WebAdminNSG rule is highlighting the fact that this rule allows inbound traffic from any external IP. In production you would create more restrictive rules.

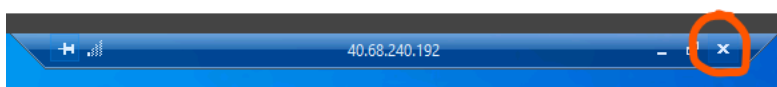


Task 4. Create Virtual Machine with Virtual Nic and Public IP

1. You wrote the code to create a virtual machine with virtual nic in Lab2 so let's re-use and add a static public IP. This code has been pre-provisioned for lab brevity.
2. In main.tf remove the line with `"*/"` found at the beginning and the line with `"\"` found at the end of the Task 4 code block. This uncomments the entire block.
3. **Note.** Address any resource reference errors if your resource identifiers do not match those expected.
4. Deploy these 3 new resources using **terraform apply**
5. **Note.** If you receive errors regarding the machine size being used ("Standard_B2s") then try "Standard_D2s_V3", "Standard_DS2_V2" or "Standard_DS3_v2".

Task 5. Testing

1. Return to the Azure Portal
2. Open the Resource groups dashboard and click on **RG3**
3. Review the list of resources and then click on virtual machine **VM3**
4. From the top menu click **Connect > Connect > Download RDP**
5. Download the RDP file, as prompted, and open
6. Connect & Accept the certificate validation and when prompted, login with your credentials from the terraform file (VM configuration) **Note:** the virtual machine may take a little while to become ready to accept an RDP connection
7. Log in using **TFadminuser/TFP@\$w0rd1234!**
8. Having verified connectivity you can disconnect for the virtual machine without waiting for it to fully initialize...



Task 6. Lab Clean-up

1. Run **terraform destroy** and confirm with **yes** when prompted.

****** Congratulations, you have completed this lab ******